

Software Engineering and Data Science
SEIS 766: Vision AI
Assignment #9

Due Date: April 16th

In this assignment you will build two neural networks using just dense layers, and train the networks based on a **contrastive loss** function that you are required to create.

1. The first network and the second network input the image embedding vectors (***ZI***) and the text embedding vectors (***ZT***) from “VAI_HW_CLIP_Img_Vecs.csv” and “VAI_HW_CLIP_Text_Vecs.csv”, respectively.
2. The two .csv files or .mat files can be found in the zip file “Image_Text_Z.zip”.
3. Both networks use dense (fully connected) layers to transform input vectors ***ZI*** and ***ZT*** to other vectors ***CZI*** and ***CZT***, respectively.
4. Implement a contrastive loss function based on the pseudo code in the “CLIP” slides.
5. After training, display the cosine similarity matrix between all the records in ***CZI*** and ***CZT***. Make sure all the vectors are sorted based on their input order (i.e. digit 0, 1, and 2). Please display your cosine similarity matrix as a color heat map and include the heatmap in your WORD document.
6. What patterns do you see in your cosine similarity matrix?
7. You can create your programs based on the sample programs under “[Course Examples + Data](#)” folder. Inside that folder, samples programs are under two subfolders: “VAI_Slide_CLIP\CelIDNA_Classify_Example\” and “\VAI_Slide_CLIP\VAI_HW_CLIP_Incomplete_PRGs\”.
8. Discussion and programming help among team members or classmates are allowed for this assignment.

Submission Guideline:

1. Please include the answers to the above questions in a WORD document.
2. Please print your programs (matlab or python) as **PDF** and include the **PDF** in your submission.
3. Please include *****ALL***** your programs a9.m/.mlx/.py/.inpyb in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.
5. Please carefully follow the submission guideline. Otherwise, the instructor may not be able to grade your assignment.